No. of Functional Features Included in the Solution

Date	25 September 2026
Team ID	SWTID-2026-4307
Project Name	AI Weather Wise API
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks
1	Real-Time Weather Aggregation	Fetches live temperature, humidity, wind, and precipitation data.	Weather Engine / OpenWeather API	Done	1.0
2	AI Travel Advice Generator	Generates natural language outdoor activity recommendations using LLMs.	AI Inference Core / OpenAI SDK	Done	1.0
3	JWT User Authentication	Secures API endpoints with bearer token validation and user registration.	Security / Auth Gateway	Done	0.5
4	Redis Caching System	Caches repeated weather queries to reduce latency and API calls.	Database / Redis Cache Manager	Done	0.5
5	Interactive OpenAPI Docs	Provides interactive Swagger UI for testing endpoints.	User Interface / FastAPI Swagger	Done	0.5
6	Predictive Weather Alerts	Sends automated weather condition anomaly alerts.	Backend / Alert Dispatcher	Done	0.5
7	Rate-Limiting Middleware	Prevents abuse by throttling request counts per API key.	Security / Middleware Layer	Done	0.5
8	Structured JSON Payloads	Returns uniform, clean JSON schemas across all REST routes.	API / Response Formatter	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Swagger UI Documentation / Interactive API Sandbox
2	Backend / Logic	3	AI Recommendations, Weather Fetcher, Alert Dispatcher
3	Database / Storage	1	Redis Caching Layer for Weather Query Storage
4	API / Integration	2	OpenWeatherMap API Integration, REST Endpoints
5	Security / Authentication	1	JWT Token Auth & Middleware Rate Limiting